**Linear Factor Model and Pricing Implications: Evidence from US and European Indices**

Oscar Segat

## 1.0: Introduction

The objective of this report is to investigate whether stock index returns are consistent with weak-form market efficiency and whether a simple linear factor model adequately explains cross-market returns. To this end, we analyse daily log returns from the S&P 500 and the EUROSTOXX 50 over a sample period of 20 years.

We first examine return dynamics and test for serial dependence using autocorrelations and variance ratio tests. We then estimate a linear factor model to assess the relationship between the two markets and discuss the implications of the results in light of the Efficient Market Hypothesis, the Arbitrage Pricing Theory, and the joint hypothesis problem.

## 1.1: Return Dynamic

The first step we operate in order to get a better understanding of the relationship between the theory of the EMH and real-world data is to analyse price data from both Europe and the USA. We use daily log returns for the S&P 500 and its European counterpart, the EUROSTOXX 50. As a first step, we compute the main summary statistics, namely the mean, variance, skewness, and kurtosis, in order to obtain a preliminary understanding of the distributional properties of returns.

Given the daily frequency of the data, the average returns are extremely small, although positive for both indices. It is worth noting that the mean return of the S&P 500 is somewhat higher than that of the EUROSTOXX 50. In contrast, the variance is relatively large compared to the mean, reflecting the substantial level of volatility typically observed in financial markets.

This high variability can be partly explained by the sequence of major economic and financial crises that have occurred since the early 2000s, including the burst of the dot-com bubble, the global financial crisis of 2008, and more recently the economic disruptions associated with the Covid-19 pandemic. Turning to higher-order moments, the results are broadly in line with well-known empirical regularities documented in the financial econometrics literature.

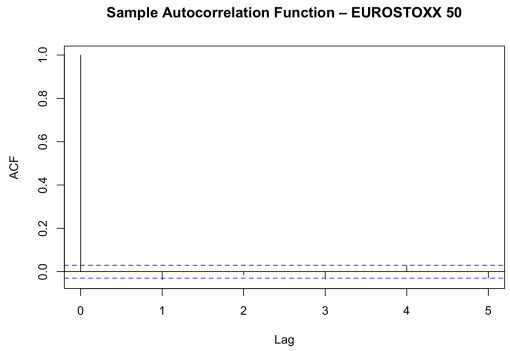
In particular, we observe negative skewness, which indicates that the distribution of returns is slightly left-skewed.

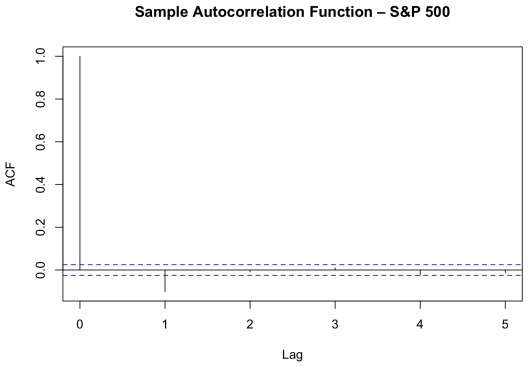
This suggests that large negative returns occur somewhat more frequently than large positive ones.

At the same time, kurtosis is significantly higher than the value associated with a normal distribution, indicating the presence of fat tails. This implies that extreme return realizations occur more often than would be predicted under the assumption of normally distributed returns.

| | Mean | Std. Dev. | Skewness | Kurtosis |
|---|---|---|---|---|
| USA | 0.021% | 1.23% | -0.38 | 10.4 |
| EUROPE | 0.0039% | 1.41% | -0.29 | 7.6 |

After describing these stylized facts, we proceed to investigate weak-form market efficiency. Under the random walk hypothesis, returns should not exhibit serial dependence, meaning that past returns should not contain predictive information about future returns.
To evaluate this property, we compute the sample autocorrelations of daily log returns for lags from one to five.
The autocorrelation function shows that, generally, estimated coefficients remain within the 95% confidence bounds, indicating that none of them are statistically different from zero.
This provides preliminary evidence that returns do not display significant short-term predictability, a finding that is consistent with the weak-form Efficient Market Hypothesis.



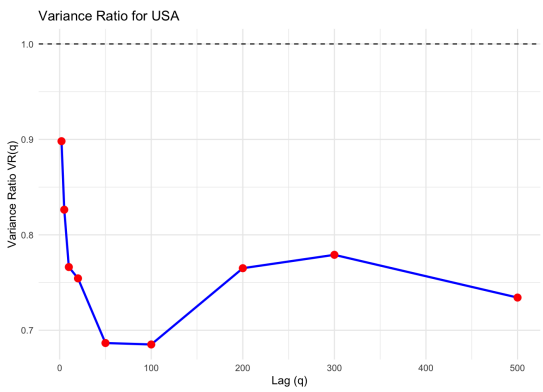

It is useful, however, to address the first (negative) spike in the autocorrelation for the S&P 500. This result is not new in literature and it has been observed by some authoritative authors[1]. Moreover it has also to be said that it could have been exaggerated from the timing we chose for the American market.
The S&P 500, in fact, starts from the first months of the year 2000, which could make the magnitude of the first lag worse than other starting points.
In Europe this is not usually observed, given the nature itself of the EUROSTOXX. The index is a union of many national indexes and this could mask possible spikes in autocorrelation, even if a tendency to a first negative spike can be seen regardless.
This short-horizon evidence motivates the use of variance ratio tests to assess return dynamics over longer horizons. We use a distance between the lags (q) of, respectively, 2, 5, 10 and 20.
We would expect, in order to further confirm the EMH, to obtain a value similar to 1. This would mean the log return process behaves like a random walk, with no tendency to regress to a mean nor to explode in volatility.
We get the following results:

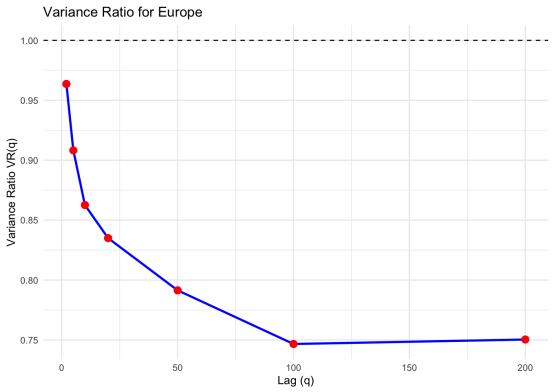
| | VR(2) | VR(5) | VR(10) | VR(20) |
|---|---|---|---|---|
| USA | 0.898 | 0.826 | 0.766 | 0.754 |
| EUROPE | 0.964 | 0.908 | 0.863 | 0.835 |

[1] Roll, R. (1984). A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market. *The Journal of Finance*, *39*(4), 1127–1139

As we can see, the sequence of the log returns behave like a random walk in the short term, but in the long term the results indicate a tendency to mean reversion.
Moreover, using the Lo-McKinlay test we get a p-value extremely low for each value of q for the S&P 500. The p-value for the EUROSTOXX 50 is, instead, higher in respect to its counterpart, but at the same time comfortably under the 5% value.
However, at the 99% confidence level, only the S&P 500 rejects the null hypothesis at all horizons, while the EUROSTOXX 50 does so only for VR(5) and VR(10).
In both cases, the negative z-statistics confirm that the deviations from the random walk are driven by mean reversion rather than momentum.

## 1.2: Long-Horizon Predictability and the Fads Model




We decided to use a higher lag for the S&P 500 index mainly because we observed an upward trend that could, eventually, bring to a Variance Ratio of 1.
For the STOXX 50, instead, we noticed a steadily decreasing variance ratio that plateaus at 0.75.
The observed difference between the two indices may stem from their structural composition. The EUROSTOXX 50, being a cross-national index, benefits from geographical diversification that smooths out idiosyncratic shocks. Conversely, the S&P 500 exhibits a sharper initial decline in the VR, likely due to higher exposure to algorithmic trading and market microstructures (e.g., bid-ask spreads) that characterize the more volatile US equity market.
In both cases we can't talk about a random walk if not in the very short term.
Economically this result suggests that we can't efficiently predict the direction of the market in the short term. In the long term, instead, the Variance Ratio suggests a significant tendency to revert to the mean.
However, this strategy of investing in the long term, expecting that the market will go back to its historical mean, already exists and it is called Value Investing, one of the most financially appreciated ways of investing.
Given the extensive dataset covering more than 20 years, the sample size is robust for short and medium horizons.
However, for longer lags, mainly for $q > 100$, we must acknowledge the potential for small-sample bias.

| | VR(50) | VR(100) | VR(200) |
|---|---|---|---|
| USA | 0.687 | 0.685 | 0.765 |
| EUROPE | 0.791 | 0.747 | 0.750 |

Even if we reduced the bias given by the usage of large lags using a sample of 20 years, this exposed our calculations to another bias: the structural bias. In fact the market structure of the market changed many times, for example with the 2008 crisis and the use of algorithmic trading in recent years.
Furthermore, the high Kurtosis found in the descriptive statistics, added to a basic knowledge of the market, suggests that there could be a presence of heteroskedasticity in the sample. This is probably given by the continuous shocks that are directly linked to the price of a stock.

1.3: Linear Factor Model and Diversification

In this section, we estimate a market model by regressing the returns of the EUROSTOXX 50 on those of the S&P 500.
This framework allows us to distinguish between common market risk and idiosyncratic risk, and to evaluate how diversification affects the overall variability of returns.

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| Alpha | -0.000137 | 0.000172 | -0.793 | 0.428 |
| Beta | 0.6735 | 0.0135 | 49.91 | <0.001 |

OLS estimation yields Alpha = −0.000137 (SE = 0.000172, t = −0.793, p = 0.428) and Beta = 0.6735 (SE = 0.0135, t = 49.91, p < 2.2e-16).
It is worth noticing that the p-value of Alpha is 0.428, which means that the probability, given that Alpha is 0, of observing a value such as the one we have observed is 42.8%.
This means that we cannot reject the hypothesis that Alpha=0.
The model achieves an R squared of 0.371, meaning that approximately 37% of the variance in EUROSTOXX 50 returns is explained by the S&P 500.
The beta coefficient is highly significant and substantially below 1, suggesting that European returns move in the same direction as US returns but with lower sensitivity.
The residual variance, instead, has a value of 0.01254%, representing the idiosyncratic component of European returns not explained by the US market factor.
Constructing an equally weighted portfolio of the two indices yields a portfolio variance of 0.01456%, which is meaningfully lower than the simple average of the two individual variances (0.01750%).
This reduction, approximately 17%, illustrates the core principle of diversification: combining assets with imperfect correlation reduces total portfolio risk.
However, it is important to note that diversification eliminates only idiosyncratic risk. The systematic component, captured by β, reflects exposure to common market movements that cannot be diversified away.

1.4 Pricing Implications

Based on the econometrics results made until now, we can make some reflections on some of the main pricing theories and problems of theoretical finance.

For what regards the main focus of this essay, the EMH, we had a potential confirmation of the model from the sample autocorrelation of the daily log returns.

After having computed the results of the variance ratios for higher lags, however, it seems likely that the main feature that drove the first result was the usage of short lags and it is probable that using higher lags we would have found higher ACF values.

The variance ratios, instead, are more useful given that they span for a larger period of time.

The results suggest that, while we cannot reject the EMH, we can advance some doubts regarding its theory.

In fact, even in its weakest form, the EMH suppose we could not gain a premium by using systematic market inefficiencies.

The variance ratio results, showing values different from 1 at higher horizons, suggest the presence of some degree of predictability, which is more consistent with a rejection of the weak form than a confirmation of it.

We have, instead, some evidence in favour of the APT.

The main proof we found is that the value of Alpha is likely equal, or very close, to 0. This means there's no premium for investing in a specific market in respect to another one.

Moreover the Beta is highly significant, this means that there's a statistically very significant sensitivity between the markets.

Both these things suggest that there's a firm link between the S&P and the EUROSTOXX, that can be interpreted as the absence of arbitrage opportunities. This would be the main presupposition of the APT theory.

It must be noticed, however, that this link is between two of the most (if not the most) liquid and efficient markets of the world.

The absence of obvious arbitrage opportunities, in this context, is near granted by the nature of the market itself.

Clearly, however, any point made in this report is subject to the Joint Hypothesis problem.

Every time we test for a determined market characteristic we are testing at the same time for the goodness of the model.

This problem is always present in any scientific research that uses complex data and cannot be in any way eliminated.

Thus, we cannot know if the results we have found are a consequence of the true structure of the market or only of the incorrect usage of the model that had been chosen.

However, being this a problem that can't be resolved, we base the results of the scientific research in general with the implicit supposition that the model is correct for the calculation of the problem.

So, in light of the results found, no economically significant abnormal returns appear to be present. The estimated alpha, while slightly negative, is statistically indistinguishable from zero and economically negligible in magnitude making any practical exploitation infeasible.

## APPENDIX

Robustness check

To further assess the robustness of the results we now re-estimate the conclusion of the paper with attention to the 2008 crisis.
Given the magnitude of the crisis, in fact, this event could have skewed the results and thus the conclusion of the research itself.
We therefore restrict the analysed period from 2010 to 2024. We then re-compute the values of the summary statistics, the variance ratios, and the OLS estimates.
The focus of this robustness check will be the S&P 500 index. The USA, in fact, are both the country where the crisis started and, consequently, the one most hit. It is then realistic to suppose that, given a low impact on results from the USA main index, the EUROSTOXX 50 would be similarly unaffected by the crisis.
We get the following results:

| | Value with 2008 crisis | Value after 2010 |
|---|---|---|
| Mean | 0.021% | 0.043% |
| Standard Deviation | 1.23% | 1.1% |
| Skewness | -0.38 | -0.72 |
| Kurtosis | 10.4 | 13.3 |
| VR(2) | 0.898 | 0.882 |
| VR(5) | 0.826 | 0.857 |
| VR(10) | 0.766 | 0.800 |
| VR(20) | 0.754 | 0.777 |
| Intercept (OLS) | -0.000137 | -0.000173 |
| Beta coefficient (OLS) | 0.6735 | 0.7257 |

As the table shows, the main differences lay both in the mean and the skewness of the newly built distribution.
However, given the fact that the variance is similar irrespective of the case, the change has little effect on variance ratios and the OLS regression in general.
Thus it is safe to say that neither the 2008, nor the dot com bubble, influenced significantly the results.